# Israel Souza Ferreira

**Data Scientist | Machine Learning Engineer**
Fortaleza, Brazil --- Remote
Email: israel.souza.dev@gmail.com | Phone: +55 85 98433-2790
LinkedIn: https://www.linkedin.com/in/isrreal/ | GitHub: https://github.com/isrreal
Portfolio: https://israelsouza.speculummaius.com.br/

## Professional Summary

Computer Science graduate with experience in applied Machine Learning research, computer vision, and systems development. I run experiments, calibration, testing, workflow design, and deployment preparation, communicating evidence and limitations clearly.

## Education

**Bachelor of Science in Computer Science**
Federal University of Ceará (UFC) --- Quixadá Campus, Brazil
Dates: 2021 -- August 2026. Completed on August 19, 2026.
Undergraduate thesis: Metaheuristics and Integer Linear Programming for Triple Roman Domination in Graphs; developed in 2024--2025 and defended in February 2025.

## Technical Skills

**Languages and data:** Python, SQL, C++, Bash, Pandas, NumPy, PostgreSQL, pgvector.
**Machine Learning and computer vision:** PyTorch, scikit-learn, SciPy, OpenCV, YuNet, SFace, ONNX Runtime, OCR, embeddings, cross-validation, calibration, and ablation.
**Engineering and quality:** FastAPI, SQLAlchemy, Alembic, Docker, GitHub Actions, Git, pytest, PWA, design systems, accessibility, and operational documentation.

## Professional Experience

**Research and Development (R&D) Researcher**
Tieta.ai --- Remote, Brazil
July 2026 -- Present
Research Fellow in the Research and Innovation Training Program.

- Contribute to research software development and maintenance, including validation and automated tests.
- Automate technical activities and produce versioned documentation for reproducibility and traceability.

**Full Stack Developer and Machine Learning Engineer**
Independent consulting for an educational institution --- Remote, Brazil
March 2026 -- Present
Project: Face Clock Evoluir. Contracted system; v1.3.0-rc.3 is an institutional supervised-pilot candidate.

- Built models, FastAPI APIs, PostgreSQL/pgvector persistence, security, tests, a design system, and operational documentation for facial attendance and medical-document triage.
- Refined the kiosk with camera feedback, identification stages, accessibility, and understandable failure recovery; simplified facial-photo selection, rotation, detection validation, and profile cropping.
- Expanded administrative workflows with an institutional calendar, collective remote work, holiday transfers, hour-bank HR review, calendar-based records, and management of employees, access, and archiving.

- Added readable, filterable audit trails, documented data-retention and deletion practices, and session-safe navigation cache, consistent transitions, and an offline PWA screen.
- Implemented a seven-step institutional setup with progressive validation, delegated IT permissions, separation of local preparation from domain publication, and HTTPS on the local network.
- Prepared and recorded three release candidates; carried out the test-environment deployment, following institutional configuration, device preparation, and hands-on flow validation.

**Machine Learning Researcher**
Tieta.ai --- Remote, Brazil
April 2025 -- June 2026
CNPq RHAE Research Fellow, part-time.

- Conducted applied Machine Learning research, developing and evaluating computational solutions.
- Ran experiments and analyzed results; established reproducible development and evaluation processes.

**Teaching Assistant --- Algorithm Design and Analysis**
Federal University of Ceará --- Quixadá Campus --- Quixadá, Brazil
March 7, 2023 -- November 30, 2023

- Supported undergraduate students in algorithms and complexity analysis.

**Teaching Assistant --- Calculus**
Federal University of Ceará --- Quixadá Campus --- Quixadá, Brazil
April 2022 -- July 2022

- Supported undergraduate students with Calculus concepts and exercises.

## Selected Projects

**Triple Roman Domination in Graphs --- Undergraduate Thesis | 2024--2025**

- Implemented genetic algorithm (GA) and ant colony optimization (ACO) metaheuristics in C++, with a Python/Pyomo integer linear programming model. Proposed correcting a published formulation that accepted invalid solutions; evaluated 362 graphs with one run per instance.

Code: https://github.com/isrreal/Triple-Roman-Domination-in-graphs

**Brazilian Emergency Aid --- Data Engineering | In development**

- Implemented Pandas and asyncpg COPY chunked ingestion for a 31.6 GB public source. On subsets of up to 5 million rows, measured post-cleanup RSS growth from about 182 to 750 MiB, using fixed 100-thousand-row chunks and three measured runs.
- Observed an association between RSS and retained IDs (r approximately 0.991); correlation does not establish causality. PostgreSQL staging-based deduplication is the next experiment, with no published improvement yet.

Code: https://github.com/isrreal/AuxilioEmergencialQueries

## Languages

Portuguese: native. English: good communication; fluent technical reading of research papers and documentation.